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## Summer Internship Report

An Introduction to Algebraic Analysis  
with an Exploration of Systems of  
Linear Integro-Differential Equations

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*Work in progress*

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August 2026

# 1 Introduction

This report is the result of an internship conducted during the summer of 2026 at the Institut Mathématique de Jussieu, Paris rive gauche where I got to learn about Algebraic Analysis especially from a constructive point of view. More precisely, the goal of this report is to expose some results in the study of systems of linear partial differential equations and time delay equations using algebraic techniques including module theory and homological algebra. Our goal will be to extract intrinsic properties of the system by studying a module presented by our system over a well-chosen ring of operators (eg. the ring of partial differential operators, time delay operators ..).

We will then move on to a more general case, namely the case systems of systems of linear integro-differential equations. The ring of integro-differential operators poses a number of challenges since it is non noetherian and the set of regular elements doesn't verify the ORE property. It is however a coherent ring which will allow us to construct free resolutions and apply homological algebra techniques. We expose a number of questions that arise from the study of this ring. Throughout the exposition of the first part, we will keep track of the hypothesis which will not be met in the case of integro differential operators. We will only provide proofs for results where such subtleties appear and will indicate references for details concerning other results.

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## 2 Studying systems of linear differential equations using algebraic tools

### 2.1 Linear systems and finitely presented D-modules

Let's start by introducing the context : let  $D$  be a ring,  $R \in D^{p \times q}$ , and  $\mathcal{F}$  a left  $D$ -module. The matrix  $R$  defines a linear system of equations. Viewing  $\mathcal{F}$  as a functional space, we want to study the solutions of  $R\eta = 0$  in  $\mathcal{F}$  ie the set

$$\ker_{\mathcal{F}}(R.) = \{\eta \in D^{p \times 1} \mid R\eta = 0\}$$

We define the left  $D$ -module,  $M \stackrel{\text{def}}{=} D^{1 \times p} / D^{1 \times q} R$ . This module plays an important role in the study of the system defined by  $R$  :

In fact, a remark due to Malgrange in [6] reveals that

$$\text{Hom}_D(M, \mathcal{F}) \cong \ker_{\mathcal{F}}(R.)$$

given by

$$\phi : f \in \text{Hom}_D(M, \mathcal{F}) \mapsto (f(y_1), \dots, f(y_p))$$

where we denote by  $\{e_i\}$  the canonical basis of  $D^{1 \times p}$ ,  $\pi$  is the projection on  $M$ , and  $y_i = \pi(e_i)$ . The inverse map is given by

$$\phi^{-1} : \eta \mapsto (\pi(\lambda) \mapsto \lambda\eta)$$

**Remark 2.1.** • *About the ring  $D$  : In much of the rest, the ring  $D$  will be a skew polynomial ring of operators defined as follows : given a ring  $A$  (typically the ring  $k[X_1, \dots, X_n]$  with  $k$  a field of characteristic zero), an  $A$  morphism  $\alpha$  and a  $\alpha$ -derivation  $\delta$  ie  $\delta(ab) = \alpha(a)\delta(b) + \delta(a)b$  for  $a, b \in A$ , we construct a non commutative polynomial ring in a formal variable  $X$  by defining  $Xa = aX + \delta(a)$  for  $a \in A$ . For example, for  $A = k[t]$ ,  $\partial = \text{id}_A$  and  $\delta$  the polynomial derivation, we get, formally, a ring of polynomials in  $X$  such that  $Xp = pX + p'$  for  $p \in A$  which corresponds to the Leibniz relation. We denote such a ring  $A[X, \alpha, \delta]$ . We denote this ring  $A < \partial >$  (the formal variable  $X$  is denoted by the symbol  $\partial$  which acts like a derivation over polynomials).*

*A similar construction with  $A = k[x_1, \dots, x_n]$  and  $n$  dérivations yields the ring of PD operators ie elements of  $D$  are polynomials in  $\partial_i$  for  $i = 1, \dots, n$  with coefficients in  $A$  where*

$$\partial_i \partial_j = \partial_j \partial_i \text{ but } \partial_j a = (\partial_j a) + a \partial_j$$

*We denote this ring  $A < \partial_1, \dots, \partial_n >$  and call it the first Weyl Algebra  $A_n$ .*

- *Notice that given the left modules  $M$  and  $\mathcal{F}$ , the set  $\text{Hom}_D(M, \mathcal{F})$  generally only inherits of a group structure and not of a module structure.*

In the rest, we will keep the following notations :  $D$  will always be a ring (it will be precised when  $D$  is supposed noetherian or commutative),  $M \in D^{p \times q}$  and  $M = \text{coker}(.R)$ .

### 2.2 The extension functor and extension groups

Malgrange's remark above motivates the study of the group  $\text{Hom}_D(M, \mathcal{F})$ . For this purpose, we introduce the following algebraic tools.

**Definition 2.2.** *Let  $D$  be a ring.*

- A complex of left (resp. right)  $D$ -modules:

$$\mathbf{M}: \cdots \xrightarrow{d_{i+2}} M_{i+1} \xrightarrow{d_{i+1}} M_i \xrightarrow{d_i} M_{i-1} \xrightarrow{d_{i-1}} \cdots$$

is a sequence of left (resp. right)  $D$ -modules where  $\text{im}(d_{i+1}) \subseteq \ker(d_i)$  for all  $i \in \mathbb{Z}$ . In other terms,  $d_i \circ d_{i+1} = 0$

- The defect of exactness of the above complex at  $M_i$  is the group  $\ker(d_i)/\text{im}(d_{i+1})$ .
- A complex of modules is said to be exact at  $M_i$  if the defect of exactness at  $M_i$  is trivial.
- Let  $M$  be a left  $D$ -module. A free (resp. projective) resolution of  $M$  is an exact sequence of the form:

$$\cdots \xrightarrow{r_2} F_1 \xrightarrow{r_1} F_0 \xrightarrow{\pi} M \rightarrow 0.$$

where  $F_i$  are free (resp. projective).

Let's go back to the situation at hand ie the study of the left  $D$ -module  $M$  finitely presented by a matrix  $R$ . If the ring  $D$  is noetherian, we can construct a free presentation of  $M$

$$\cdots D^{1 \times r} \xrightarrow{R_2} D^{1 \times p} \xrightarrow{R} D^{1 \times q} \xrightarrow{\pi} M \rightarrow 0 \quad (1)$$

Given a left  $D$ -module  $\mathcal{F}$ , we apply the contravariant functor  $\text{Hom}(-, \mathcal{F})$  to (1), and using the facts that  $\text{Hom}(D^{1 \times a}, \mathcal{F}) \cong \mathcal{F}^a$  for all  $a \in \mathbb{N}$  and  $\text{Hom}(-, \mathcal{F})(R)(\eta) = R\eta$ , we get the following complex :

$$\cdots \mathcal{F}^r \xleftarrow{R_2} \mathcal{F}^p \xleftarrow{R} \mathcal{F}^q \leftarrow 0 \quad (2)$$

The defects of exactness of (2) are called the extensions groups of  $M$  by  $\mathcal{F}$  denoted  $\text{ext}_D^i(M, \mathcal{F})$ . If we write  $R = R_1$ , we have

$$\begin{cases} \text{ext}_D^0(M, \mathcal{F}) = \ker_{\mathcal{F}}(R_1) \cong \text{Hom}_D(M, \mathcal{F}) \\ \text{ext}_D^i(M, \mathcal{F}) = \ker((R_{i+1} \cdot) / \text{im}(R_i \cdot)) \end{cases}$$

**Remark 2.3.** • A classical result in homological algebra tells us that the extension groups do not depend on the specific resolution in (1). They are entirely determined by  $M$  and  $\mathcal{F}$ . (cf [9] for details).

- If we consider the inhomogenous equation  $Rx = \eta$ , then a necessary condition for the system to have a solution is  $R_2\eta = 0$  since (1) is exact. We call the condition  $R_2\eta = 0$  the compatibility condition of the inhomogenous system. This condition is sufficient iff  $\text{ext}_D^1(M, \mathcal{F}) = 0$ . Hence,  $\text{ext}_D^1(M, \mathcal{F})$  indicates "how close the compatibility condition is to being sufficient".

At this point, a natural question to ask is which information about  $M$  can we extract from computing the extension groups. Let's start by some classical definitions :

**Definition 2.4.** (free, stably free, projective, reflexive, torsion-free modules)

Let  $D$  be a ring and  $M$  a left (resp. right)  $D$ -module.

- $M$  is a free module if there exists  $n \in \mathbb{N}$  such that  $M \cong D^n$ .
- $M$  is stably free if there exists  $r, n \in \mathbb{N}$  such that  $M \oplus D^r \cong D^n$
- $M$  is projective if there exists a left (resp. right)  $D$ -module  $N$  and  $n \in \mathbb{N}$  such that  $M \oplus N \cong D^n$ . Equivalently,  $M$  is projective if for any  $D$ -module  $N$ ,  $\text{ext}_D^i(M, N) = 0 \forall i \in \mathbb{N}$ .
- $M$  is reflexive if the canonical morphism :

$$\epsilon : M \rightarrow M^{**}$$

$$m \mapsto (f \mapsto f(m))$$

is an isomorphism, where  $M^{**} = \text{Hom}_D(\text{Hom}_D(M, D), D)$ .

- If  $D$  is a noetherian domain : the torsion submodule of  $M$  is the set :

$$t(M) \stackrel{\text{def}}{=} \{m \in M \mid \exists d \in D \setminus \{0\} \, dm = 0\}$$

- If  $t(M) = 0$  then  $M$  is said to be torsion-free.

**Remark 2.5.** For the definition of the torsion submodule, noetherianity is actually not needed. It is sufficient to have the following property, called the ORE property :

**Definition 2.6.** (ORE property) A multiplicative subset  $S \subset D$  verifies the ORE property if  $\forall d \in D$  and  $s \in S$ , there exists  $d' \in D \setminus \{0\}$  and  $s' \in S \setminus \{0\}$  such that  $ds' = sd'$

The fact that  $t(M)$  is a left  $D$ -submodule of  $M$  is a consequence of the left Ore property of  $D$  (which comes from the left noetherian domain property). Indeed, for all  $m_1, m_2 \in t(M)$  and all  $d_1, d_2 \in D$ , we need to prove that  $d_1m_1 + d_2m_2 \in t(M)$ . Since  $m_1, m_2 \in t(M)$ , there exist  $p_1, p_2 \in D \setminus \{0\}$  such that  $p_1m_1 = 0$  and  $p_2m_2 = 0$ . Using the left Ore property of  $D$ , there exist non-trivial  $r_1, r_2, s_1, s_2, t_1, t_2 \in D$  satisfying:

$$r_1p_1 = s_1d_1, \quad r_2p_2 = s_2d_2, \quad t_1s_1 = t_2s_2.$$

Therefore, we get

$$(t_1s_1)(d_1m_1 + d_2m_2) = t_1(s_1d_1)m_1 + t_2(s_2d_2)m_2 = t_1r_1(p_1m_1) + t_2r_2(p_2m_2) = 0,$$

which shows that  $d_1m_1 + d_2m_2 \in t(M)$  since  $t_1s_1 \in D \setminus \{0\}$ . We will come back to the study of torsion elements in non noetherian rings in the second part of this report when we will dive into integro differential equations.

**Definition 2.7.** (The Auslander transpose)

Given a left  $D$ -module  $M = D^{1 \times q} / D^{1 \times p} R$  finitely presented by a matrix  $R$ , the Auslander transpose of  $M$  is the right  $D$ -module,  $D^p / RD^q$ .

**Proposition 2.8.** ([9]) We have the following implications :

$$\text{free} \implies \text{stably free} \implies \text{projective} \implies \text{reflexive} \implies \text{torsion-free}$$

The following theorems establish the utility of computing extension groups :

**Theorem 2.9.** (i think noeth is needed here) Let  $M$  be a finitely presented  $D$ -module. If  $\text{ext}_D^1(M, D) = 0$  then  $M$  is a torsion  $D$ -module.

**Theorem 2.10.** [7] Let  $D$  be a noetherian domain. We further suppose that  $D$  is either commutative or ORE. With the same notations as above, we have the following isomorphism

$$t(M) \cong \text{ext}_D^1(N, D)$$

hence  $M$  is torsion-free iff  $\text{ext}_D^1(N, D) = 0$ .

*Proof.* In the first part of the proof, we establish the following diagram : given  $D$ -modules  $M$  and  $N$  as defined above, and a short exact sequence of right  $D$ -modules

$$0 \rightarrow X \rightarrow Y \rightarrow Z \rightarrow 0$$

$$\begin{array}{ccccccc}
& & & & & \vdots & \\
& & & & & \downarrow & \\
& & & & & \text{Tor}_1^A(X, M) & \\
& & & & & \downarrow & \\
& & & & & \text{Tor}_1^A(Y, M) & \\
& & & & & \downarrow & \\
& & & & & \text{Tor}_1^A(Z, M) & \\
& & & & & \downarrow & \\
& & 0 & 0 & 0 & & \\
& & \downarrow & \downarrow & \downarrow & & \\
0 \rightarrow & \text{Hom}_A(N, X) & \rightarrow & X \otimes_A F_1 & \rightarrow & X \otimes_A F_0 & \rightarrow & X \otimes_A M & \rightarrow & 0 \\
& \downarrow & & \downarrow & & \downarrow & & \downarrow & & \\
0 \rightarrow & \text{Hom}_A(N, Y) & \rightarrow & Y \otimes_A F_1 & \rightarrow & Y \otimes_A F_0 & \rightarrow & Y \otimes_A M & \rightarrow & 0 \\
& \downarrow & & \downarrow & & \downarrow & & \downarrow & & \\
0 \rightarrow & \text{Hom}_A(N, Z) & \rightarrow & Z \otimes_A F_1 & \rightarrow & Z \otimes_A F_0 & \rightarrow & Z \otimes_A M & \rightarrow & 0 \\
& \downarrow & & \downarrow & & \downarrow & & \downarrow & & \\
& \text{ext}_A^1(N, X) & & 0 & & 0 & & 0 & & \\
& \downarrow & & & & & & & & \\
& \text{ext}_A^1(N, Y) & & & & & & & & \\
& \downarrow & & & & & & & & \\
& \text{ext}_A^1(N, Z) & & & & & & & & \\
& \downarrow & & & & & & & & \\
& \vdots & & & & & & & &
\end{array} \tag{3}$$

0

This diagram does not need the assumptions on  $D$ . We then apply this diagram to the following exact sequence :  $0 \rightarrow A \rightarrow K \rightarrow K/A \rightarrow 0$  where  $K$  is the field of quotients of  $D$ . The existence of  $K$  is not given for an arbitrary ring, this is where the additional assumptions come in place. We then obtain the following diagram :

$$\begin{array}{ccccccc}
& & & & & 0 & \\
& & & & & \downarrow & \\
0 \rightarrow & \text{Hom}_A(N, A) & \rightarrow & F_1 & \rightarrow & F_0 & \rightarrow & M & \rightarrow & 0 \\
& \downarrow & & \downarrow & & \downarrow & & \downarrow & & \\
0 \rightarrow & \text{Hom}_A(N, K) & \rightarrow & K \otimes_A F_1 & \rightarrow & K \otimes_A F_0 & \rightarrow & K \otimes_A M & \rightarrow & 0 \\
& \downarrow & & \downarrow & & \downarrow & & \downarrow & & \\
0 \rightarrow & \text{Hom}_A(N, K/A) & \rightarrow & (K/A) \otimes_A F_1 & \rightarrow & (K/A) \otimes_A F_0 & \rightarrow & (K/A) \otimes_A M & \rightarrow & 0 \\
& \downarrow & & \downarrow & & \downarrow & & \downarrow & & \\
& \text{ext}_A^1(N, A) & & 0 & & 0 & & 0 & & \\
& \downarrow & & & & & & & & \\
& 0 & & & & & & & &
\end{array} \tag{4}$$

where we have used that  $K$  is a flat module (ie the functor  $- \otimes K$  is exact) and that  $\text{ext}_D^1(N, K) \cong K \otimes \text{ext}_D^1(N, A)$ . The snake lemma applied to this diagram then concludes the proof.  $\square$

**Remark 2.11.** *The proof above relies on the existence of the field of fractions ie the fact that it is possible to take the localization of  $D$  with respect to its regular elements which supposes that the set of regular elements verifies the ORE property. We will see later that this is not the case for the ring of integro-differential operators and thus this proof fails in that case.*

**Definition 2.12.** *(The projective and global dimensions) Given a projective resolution,*

$$\cdots \rightarrow P_2 \rightarrow P_1 \rightarrow P_0 \rightarrow M \rightarrow 0$$

*we say that the resolution is of length  $n \in \mathbb{N}$  if there exists  $n$  such that  $P_m = 0$  for all  $m \geq n + 1$ .*

- *The left (resp. right) projective dimension of a left (resp. right)  $D$ -module  $M$  is the minimum of the length over the projective resolutions of  $M$ . It is denoted  $\text{lpd}_D(M)$  (resp  $\text{rpd}_D(M)$ ) and is set to be equal to  $\infty$  if there exists no resolution of finite length.*
- *The left (resp. right) global dimension of a ring  $D$  is the supremum of  $\text{lpd}_D(M)$  over the left (resp. right)  $D$ -modules.*
- *If the left and right global dimensions of a ring  $D$  coincide, we call the common value the global dimension of  $D$ .*

**Theorem 2.13.** • *We have the following exact sequence :*

$$0 \rightarrow \text{ext}_D^1(N, D) \rightarrow M \xrightarrow{\epsilon} M^{**} \rightarrow \text{ext}_D^2(N, D) \rightarrow 0$$

- *$M$  is reflexive iff  $\text{ext}_D^i(N, D) = 0$  for  $i = 1, 2$ .*
- *$M$  is projective iff  $\text{ext}_D^i(N, D) = 0$  for  $i = 1, \dots, \text{gld}(D)$ .*

Now that we have the tools to study the left  $D$ -module  $M$  independantly of the free-resolution chosen, we establish further relations between the algebraic properties of  $M$  and some properties of the system given by the matrix  $R$ .

## 2.3 Parametrizability and module properties

A classical question in control theory is whether it is possible to transport the state variable from one state to another by controlling the inputs. Intuitively, this depends on whether the state variables are bound by other constraints than the inputs. This leads to the following definitions :

**Definition 2.14.** *(Observables, autonomous observables, parametrizability) Given a ring of operators  $D$  and a system of the form  $R\eta = 0$  where  $R \in D^{p \times q}$ , we call :*

- *an observable of the system, any  $D$ -linear combination of  $\eta_i$  where  $\eta$  is a solution.*
- *an observable  $\phi(\eta)$  is said to be autonomous if it satisfies a non trivial equation over  $D$  ie if there exists  $d \in D \setminus \{0\}$  such that  $d\phi(\eta) = 0$ .*
- *a system is said to be autonomous if all of its observables are autonomous.*
- *A system is parametrizable if there exists  $Q \in D^{q \times r}$  such that :*

$$\{\eta \in \mathcal{F}^q \mid R\eta = 0\} = \{Qx \mid x \in \mathcal{F}^r\}$$

*In this case, the matrix  $Q$  determines a parametrization of the system.*

**Theorem 2.15.** [4] *A system is autonomous iff it is parametrizable.*

The goal of this section is thus to understand when the system is parametrizable.

**Corollary 2.16.** *of theorems 2.16 and 2.13 :*

*Let  $D$  be a noetherian domain with finite global dimension  $\text{gld}(D)$ ,  $R \in D^{p \times q}$  and  $M$  the left  $D$ -module finitely presented by  $R$ . Denoting  $R$  by  $Q_1$ , we have*

- *$M$  is torsion-free iff there exists  $Q_2 \in D^{r_2 \times p}$  such that the following sequence is exact :*

$$D^{q \times 1} \xrightarrow{\cdot Q_1} D^{p \times 1} \xrightarrow{\cdot Q_2} D^{r_2 \times 1} \quad (5)$$

- $M$  is reflexive iff there exists  $Q_2 \in D^{r_1 \times p}$  and  $Q_3 \in D^{r_3 \times r_2}$  such that the following sequence is exact :

$$D^{q \times 1} \xrightarrow{\cdot Q_1} D^{p \times 1} \xrightarrow{\cdot Q_2} D^{r_2 \times 1} \xrightarrow{\cdot Q_3} D^{r_3 \times 1} \quad (6)$$

- $M$  is projective iff there exists  $Q_2 \in D^{r_1 \times p}$  and  $Q_i \in D^{r_i \times r_{i-1}}$  for  $i = 3, \dots, \text{gld}(D)$  such that the following sequence is exact :

$$D^{q \times 1} \xrightarrow{\cdot Q_1} D^{p \times 1} \xrightarrow{\cdot Q_2} D^{r_2 \times 1} \xrightarrow{\cdot Q_3} D^{r_3 \times 1} \dots \xrightarrow{\cdot Q_{\text{gld}(D)}} D^{\text{gld}(D) \times 1} \quad (7)$$

**Example 2.17.** Let's take a look at a classic example, namely that of the curl, divergence and gradient operators.

Now, given a left  $D$ -module  $\mathcal{F}$ , we can apply the functor  $\text{hom}(-, \mathcal{F})$  to (5), (6). In the case (5) for example, we get

$$\mathcal{F}^q \xleftarrow{R} \mathcal{F}^p \xleftarrow{Q_2} \mathcal{F}^{r_2}$$

and the defect of exactness at  $\mathcal{F}^p$  measures how far the matrix  $Q_2$  is of parametrizing the solution space. We are interested in the case where the obtained complexes are exact sequences, hence the following definition :

**Definition 2.18.** (Injective modules)

Let  $D$  be a ring and  $\mathcal{F}$  a  $D$ -module.  $\mathcal{F}$  is injective if for all  $D$ -modules  $N$ , we have  $\text{ext}_D^i(N, \mathcal{F}) = 0$  for all  $i \in \mathbb{Z}$ .

**Example 2.19.** If  $k$  is a field of characteristic 0,  $D = k[\partial_1, \dots, \partial_n]$  and  $\Omega$  an open convex set of  $\mathbb{R}^n$ , then the  $D$ -modules  $C^\infty(\Omega)$  (resp.  $\mathcal{D}'(\Omega), \mathcal{S}'(\Omega), \mathcal{A}(\Omega), \mathcal{B}(\Omega)$ ) of smooth functions (resp. distributions, temperate distributions, real analytic functions, hyperfunctions) are injective.

So we get the following corollary to corollary 2.16 :

**Corollary 2.20.** Taking the same hypotheses as in 2.16 and letting  $\mathcal{F}$  be an injective left  $D$ -module, we obtain necessary and sufficient conditions on the module  $M$  for  $\ker_{\mathcal{F}}(R)$  to be parametrized.

## 2.4 Baer extensions and Monge parametrization

### 2.4.1 Baer extensions

In the previous sections, we studied algebraic properties of the finitely presented left module  $M$  and established relations between  $M$  and the parametrizability of the system. We now know that the system is parametrizable iff  $M$  is torsion-free and we can compute a potential  $Q$ . A natural question is then the following : given a module  $M$  finitely presented by  $R$ , can we isolate  $t(M)$  and  $M/t(M)$  to take advantage of the parametrizability ? Another way to express this is : can we decouple the parametrizable and non parametrizable parts, solve them separately while preserving global information about the module  $M$  ? This naturally leads to consider the set of extensions of  $t(M)$  by  $M/t(M)$  ie the set of exact sequences of the form :

$$0 \rightarrow t(M) \xrightarrow{f} E \xrightarrow{g} E/t(M) \rightarrow 0$$

Let's generalize the discussion to the extensions of a  $D$ -module  $N_1$  by a  $D$ -module  $N_2$  with the goal of applying the result to  $t(M)$  and  $M/t(M)$ . On this set, we define the equivalence relation  $\sim$  : two exact sequences

$$e : 0 \rightarrow N_1 \xrightarrow{f} E \xrightarrow{g} N_2 \rightarrow 0$$



and

$$e' : 0 \rightarrow N_1 \xrightarrow{f'} E' \xrightarrow{g'} N_2 \rightarrow 0$$

are equivalent if there exists a  $D$ -linear morphism  $h : E \rightarrow E'$  such that

$$h \circ f = f' \text{ and } g = g' \circ h$$

We denote by  $e_D(N_1, N_2) \stackrel{\text{def}}{=} \{\text{extensions of } N_1 \text{ by } N_2\} / \sim$  the set of equivalence classes. It turns out this set has an interesting structure as shown in the following theorem :

**Theorem 2.21.** [3], [5] *Given two  $D$ -modules  $N_1$  and  $N_2$ , the set  $e_D(N_1, N_2)$  has a group structure where  $[e] + [e']$  is given by the equivalence class of the following sequence : Furthermore, we have a group isomorphism :*

$$\text{ext}_D^1(N_1, N_2) \cong e_D(N_1, N_2)$$

Now let's restrict to the case of finitely presented  $D$ -modules. Let  $N_1 = \text{coker}(.R)$  and  $N_2 = \text{coker}(S)$  where  $R \in D^{p \times q}$ ,  $S \in D^{s \times t}$ . Let's consider the beginning of a free resolution of  $N_1$  :

$$\dots D^{1 \times l} \xrightarrow{.R_2} D^{1 \times p} \xrightarrow{.R} D^{1 \times q} \xrightarrow{\pi} N_1 \rightarrow 0$$

where  $\pi$  is the canonical projection. Applying the contravariant functor  $\text{Hom}_D(-, N_2)$  yields :

$$\dots N_2^l \xleftarrow{R_2} N_2^p \xleftarrow{R} N_2^q \leftarrow \text{Hom}(N_1, N_2)$$

hence  $\text{ext}_D^1(N_1, N_2) = \ker_{N_2}(R_2) / \text{im}_{N_2}(R)$ . Let's now characterize  $\text{ext}_D^1(N_1, N_2)$  : let's consider a finite representation of  $N_2$  :

$$D^{1 \times s} \xrightarrow{.R_2} D^{1 \times t} \xrightarrow{\delta} N_2 \rightarrow 0$$

Taking  $m$  copies of this sequence, we get :

$$D^{m \times s} \xrightarrow{.R_2} D^{m \times t} \xrightarrow{\text{id}_m \otimes \delta} N_2^m \rightarrow 0$$

Now taking  $m = p, q$  and  $l$ , we get the following commutative diagram :

$$\begin{array}{ccccc} & 0 & & 0 & & 0 \\ & \uparrow & & \uparrow & & \uparrow \\ N_2^l & \xleftarrow{R_2} & N_2^p & \xleftarrow{R} & N_2^q \\ \text{id}_l \otimes \delta \uparrow & & \text{id}_p \otimes \delta \uparrow & & \text{id}_q \otimes \delta \uparrow \\ D^{l \times t} & \xleftarrow{R_2} & D^{p \times t} & \xleftarrow{R} & D^{q \times t} \\ .S \uparrow & & .S \uparrow & & .S \uparrow \\ D^{l \times s} & \xleftarrow{R_2} & D^{p \times s} & \xleftarrow{R} & D^{q \times s} \end{array} \quad (8)$$

This diagram shows that :

$$\begin{aligned} \ker_{N_2}(R_2) &= \{ \text{id}_p \otimes \delta(A) \mid A \in D^{p \times t}, \text{id}_l \otimes \delta(R_2 A) = 0 \} \\ &= \{ \text{id}_p \otimes \delta(A) \mid A \in D^{p \times t}, \exists B \in D^{l \times s} \text{ st } R_2 A = BS \} \end{aligned}$$

and

$$\begin{aligned} \text{Im}_{N_2}(R) &= \{ R(\text{id}_q \otimes \delta)(A) \mid A \in D^{q \times t} \} \\ &= \{ \text{id}_p \otimes \delta(A) \mid A \in D^{p \times t} \text{ st } \exists X \in D^{q \times t}, \text{id}_p \otimes \delta(A - RX) = 0 \} \\ &= \{ \text{id}_p \otimes \delta(A) \mid A \in D^{p \times t} \text{ st } \exists X \in D^{q \times t}, B \in D^{p \times s}; A = RX + BS \} \end{aligned}$$

Denoting  $\Omega \stackrel{def}{=} \{A \in D^{p \times t} \mid \exists B \in D^{l \times s}; R_2 A = BS\}$ , we get the following isomorphism :

$$v : ext_D^1(N_1, N_2) \xrightarrow{\sim} \Omega / (RD^{q \times t} + D^{p \times s}S)$$

$$\rho(id_p \otimes \delta(A)) \mapsto \pi'(A)$$

where  $\rho : ker_{N_2}(R_2) \rightarrow ker_{N_2}(R_2)/im_{N_2}(R_2)$  and  $\pi' : \Omega \rightarrow \Omega / (RD^{q \times t} + D^{p \times s}S)$  are the canonical projections.

Putting together the isomorphism of 2.21 and the above isomorphism, we get  $e_D(N_1, N_2) \cong \Omega / (RD^{q \times t} + D^{p \times s}S)$ . Expliciting this isomorphism in [8] gives a nice representation of every equivalence class of extension :

**Theorem 2.22.** *Let  $D$  be a ring,  $S \in D^{p \times q}$ ,  $R \in D^{s \times t}$ ,  $N_1 = D^{1 \times p}/D^{1 \times q}R$  and  $N_2 = D^{1 \times s}/D^{1 \times t}S$ . Futhermore, let  $R_2$  be such that  $ker(.R) = im(.R_2)$ . Then every equivalence class of extensions of  $N_1$  by  $N_2$  is represented by an extension of the form*

$$e : 0 \rightarrow N_1 \xrightarrow{\alpha} E \xrightarrow{\beta} N_2 \rightarrow 0$$

where  $E = coker(.Q)$  with  $Q = \begin{pmatrix} S & -A \\ 0 & R \end{pmatrix}$ . Moreover, the equivalence class of the extension only depends on the equivalence class of  $A$  in  $\Omega / (RD^{q \times t} + D^{p \times s}S)$  and  $\Omega = \{A \in D^{p \times t} \mid \exists B \in D^{l \times s}; R_2 A = BS\}$

Going back to our primary goal of studying the extension of  $t(M)$  by  $M/t(M)$ , let's show that we are indeed in the case of two finitely presented matrices. We know that if

$$\dots D^r \xrightarrow{Q} D^q \xrightarrow{R} D^p \xrightarrow{\pi'} N \rightarrow 0$$

is a free presentation of  $N$ , the Auslander transpose of  $M$  then taking the functor  $Hom(-, D)$ , we get  $ext_D^1(N, D) \cong ker_D(.Q)/im_D(.R)$ . Now **if  $D$  is a noetherian ring**, we can find a matrix  $R' \in D^{p' \times q}$  for some  $p' \in \mathbb{N}$  such that  $ker_D(.Q) = im_D(.R')$  so

$$t(M) \cong ext_D^1(M, D) = im_D(.R')/im_D(.R)$$

The following lemma shows that in this case, the torsion submodule is in fact finitely presented :

**Lemme 2.23.** *Let  $R \in D^{p \times q}$ ,  $R' \in D^{p' \times q}$  such that  $im(.R) \subseteq im(.R')$ . Let  $R'' \in D^{p \times p'}$  such that  $R = R''R'$ . Moreover, let  $R_2 \in D^{q' \times p'}$  for some  $q' \in \mathbb{N}$  such that  $ker_D(.R') = im_D(.R_2)$  then*

$$im_D(.R')/im_D(.R) \cong D^{1 \times p'} / (D^{1 \times q'} R_2 + D^{1 \times q} R'') = coker_D(. \begin{pmatrix} R_2 \\ R'' \end{pmatrix})$$

So  $t(M)$  is finitely presented, let's call  $T$  the matrix  $\begin{pmatrix} R_2 \\ R'' \end{pmatrix}$  such that  $t(M) = coker(.T)$ .

Moreover,

$$M/t(M) \cong \frac{coker(.R)}{im_D(.R')/im_D(.R)} \cong coker(.R')$$

is also finitely generated and we can apply 2.22 with  $R = \begin{pmatrix} R_2 \\ R'' \end{pmatrix}$  and  $S = R'$ .

But it turns out we can find a very simple representation of the matrix  $A$  in  $\Omega / (RD^{q \times t} + D^{p \times s}S)$ .

**Theorem 2.24.** [8] Given  $D$  a noetherian ring, and matrices,  $R \in D^{p \times q}$ ,  $R' \in D^{r \times p}$ ,  $R'' \in D^{p \times p'}$  and  $R_2 \in D^{q' \times p'}$  then the exact sequences :

$$0 \rightarrow t(M) \xrightarrow{i} M \xrightarrow{\pi} M/t(M) \rightarrow 0$$

and

$$0 \rightarrow t(M) \rightarrow E \rightarrow M/t(M) \rightarrow 0$$

are equivalent where  $E = \text{coker}(.P)$  and  $P = \begin{pmatrix} R' & -\text{Id}_r \\ 0 & R_2 \\ 0 & R'' \end{pmatrix}$

In particular,  $\ker_{\mathcal{F}}(R.) \cong \text{Hom}_D(M, \mathcal{F}) \cong \text{Hom}_D(E, \mathcal{F}) \cong \ker_{\mathcal{F}}(P.)$  where the isomorphism between the solution spaces is as follows :

$$\eta \in \ker_{\mathcal{F}}(R.) \mapsto \begin{pmatrix} \text{Id}_p \\ R' \end{pmatrix} \eta \text{ and } \begin{pmatrix} \zeta \\ \theta \end{pmatrix} \mapsto \zeta$$

This leads us to solve the system :

$$\begin{cases} R'\zeta - \theta = 0, \\ R_2\theta = 0, \\ R''\theta = 0. \end{cases}$$

This system can be integrated in cascade :

- we start by solving the homogenous system  $\begin{pmatrix} R_2 \\ R'' \end{pmatrix} \theta = 0$
- then we find a particular solution  $\zeta^*$  to  $R'\zeta = \theta$ .
- Since  $M/t(M) = \text{coker}(.R')$ , is torsion free, if  $\mathcal{F}$  is an injective  $D$ -module, then by corollary 2.20, there exists  $Q \in D^{p \times n}$  such that  $\ker_{\mathcal{F}}(.R') = Q\mathcal{F}^n$ .

So a solution of our initial system is of the form  $\zeta^* + Qx$  for any  $x \in \mathcal{F}^n$ . This parametrization is called a Monge parametrization of the system  $R\zeta = 0$ .

### 3 The case of integro-differential systems

After studying systems of linear differential equations and time delays, it is natural to wonder what we can say about integro-differential systems of equations. For this we introduce the endomorphism  $I : p \in K[x] \mapsto \int_0^x p(t)dt$ . The ring denoted  $\mathbb{I}_1$  of integro differential operators is defined as the subring of  $\text{End}(K[x])$  generated by  $\partial, I$  and  $m_a \forall a \in K[x]$  where  $m_a$  is the multiplication by  $a$ .

#### 3.1 Definitions, identities and the normal form

We start by introducing some properties of  $\mathbb{I}_1$  that will be very useful for the rest.

- First we observe that, formally,  $I$  is a right inverse of  $\partial$ , but  $I\partial = 1 - e$  where  $e : p \in K[t] \mapsto p(0)$ . Thus  $\mathbb{I}_1$  is not commutative.

**Lemme 3.1.** For any element  $f \in I_1$ , there exists unique  $a_i \in K$  such that  $f =$ . This is called the normal form of  $f$ .

## 3.2 Algebraic properties of $\mathbb{I}_1$

**The involution  $\theta$  :**

**Lemme 3.2.**  $\mathbb{I}_1$  contains a unique bilateral ideal namely  $\langle e \rangle$ .

Let's study the quotient ring  $\mathbb{B}_1 \stackrel{def}{=} \mathbb{I}_1 / \langle e \rangle$ . If  $x \in \mathbb{I}_1$ , we denote by  $[x]$  its image in the quotient  $\mathbb{I}_1 / \langle e \rangle$ . Then .. Hence  $\mathbb{B}_1$  is a skewed polynomial ring over  $D$ .

### 3.2.1 $\mathbb{I}_1$ is a non noetherian coherent ring

Let  $n \in \mathbb{N}$ . We denote by  $T_n$  the  $n^{th}$  Taylor operator ie  $T_n = \sum_{i=0}^n \frac{t^i e \partial^i}{i!}$  then we have  $T_{n+1}T_n = T_n$  hence  $T_n\mathbb{I}_1 \subset T_{n+1}\mathbb{I}_1$  and the equality is strict. In fact, if there exists  $h \in \mathbb{I}_1$  such that  $T_{n+1} = T_n h$  then  $T_n(1 - T_n h) = 0$  hence  $1 - T_n h \in \langle e \rangle$  which is absurd since  $T_n h \in \langle e \rangle$  and  $\langle e \rangle$  is a proper ideal. This proves that  $\mathbb{I}_1$  is not noetherian.

**Theorem 3.3.** For every  $p, q \in \mathbb{N}$  and  $R \in \mathbb{I}_1^{p \times q}$ ,  $\ker(.R)$  and  $\ker(R.)$  are finitely generated ideals.

*Proof.* □

### 3.2.2 Torsion elements in $\mathbb{I}_1$

In the first section of this report, we used the noetherian property of the studied rings in order to construct free resolutions and study the extension groups. The coherence of  $\mathbb{I}_1$  enables us to do the same thing in this context. However, theorem 2.2 is not true in this context since the ring of quotients does not even exist in general in [1], Bavula shows that  $\mathbb{I}_1$  cannot be embedded in a ring of quotients.

**Theorem 3.4.** The regular elements of  $\mathbb{I}_1$  don't verify the ORE property. In particular,  $\mathbb{I}_1$  does not admit a ring of quotients.

*Proof.* □

Hence the result given in 2.2 does not hold.

## 3.3 The Bass module and open questions

Even though the torsion submodule is not well defined in this context, the submodule  $\ker(\epsilon) = \text{Ext}_{\mathbb{I}_1}^1(M, \mathbb{I}_1)$  is still well defined. We call it the *Bass-submodule* and will denote it  $\text{Bass}(M, \mathbb{I}_1)$

### 3.3.1 Torsion elements and the Bass submodule :

Let  $M$  be a  $\mathbb{I}_1$ -module, and  $m \in M$  a torsion element. Then  $m \in \text{Bass}(M, \mathbb{I}_1)$ . In fact,  $\text{Bass}(M, \mathbb{I}_1) = \{m \in M \mid \forall f \in M^*, f(m) = 0\}$  so given  $f \in M^*$  and  $m$  a torsion element, there exists a regular element  $a \in \mathbb{I}_1$  such that  $am = 0$  hence  $f(am) = af(m) = 0$  since  $f$  is  $\mathbb{I}_1$  linear. And since  $a$  is regular, we get  $f(m) = 0$ .

A natural question is whether there is a deeper relation between the torsion elements and the Bass-submodule. For example, in the cases where the torsion elements do form a submodule of  $M$ , do we have equality between the torsion and Bass submodules.

### 3.3.2 Interpreting $\text{Bass}(M, \mathbb{I}_1)$ :

Going back to what we developed in section 2.4, we can apply the results to  $N_1 = \text{Bass}(M, \mathbb{I}_1)$  and  $N_2 = M / \text{Bass}(M, \mathbb{I}_1)$ .

**Lemme 3.5.** [2] The global dimension of  $\mathbb{I}_1$  is equal to 1.

In particular, a reflexive  $\mathbb{I}_1$ -module is projective. Applying this to the exact sequence

$$0 \rightarrow \text{Bass}(M, \mathbb{I}_1) \xrightarrow{\epsilon} M \rightarrow M/\text{Bass}(M, \mathbb{I}_1) \rightarrow 0$$

we get  $M = \text{Bass}(M, \mathbb{I}_1) \oplus M/(\text{Bass}(M, \mathbb{I}_1))$ . We can show that in the case of a short exact sequence that splits, the matrix  $A$  given by theorem 2.24 can be chosen to be equal to 0 hence we obtain a block diagonal matrix.

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